

MATH-414 Stochastic simulation
Prof. Fabio Nobile

Project 10

Stochastic simulation of epidemic models

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Exercise (a)

We consider a population of $N = 100$ individuals and an initial condition $(S_0, I_0) = (99, 1)$. We will implement the stochastic SIR, its mean field and diffusion approximations for a time $T = 10$. We let $\beta = 0.02$ and $\gamma = 0.4$.

Stochastic SIR:

The stochastic SIR model can be modeled by the following discrete-state, continuous time Markov process $\{X(t) = (S(t), I(t)) \in \{0, 1, \dots, N\}^2,\ t > 0\}$ with transition probabilities:

$$\mathbb{P}(X(t + dt) = (s, i) + (-1, 1) \mid X(t) = (s, i)) = \beta sidt + o(dt)$$

$$\mathbb{P}(X(t + dt) = (s, i) + (0, -1) \mid X(t) = (s, i)) = \gamma idt + o(dt),$$

where $\beta, \gamma \in \mathbb{R}^+$.

This model is simulated by the following algorithm. Start with $X(0) = (S(0), I(0)) = (99, 1)$, and while $t < T$:

- generate an exponentially distributed holding time S, with parameter $(\beta S(t)I(t) + \gamma I(t))$.
- Set $t = t + S$.
- Generate $U \sim U(0, 1)$.
- Compute the probability of recovery $P = \frac{\beta S(t)I(t)}{\beta S(t)I(t)+\gamma I(t)}$.
- If $U \leq P$, set $X(t) = X(t) + (-1, 1)$.
- If $U \geq P$, set $X(t) = X(t) + (0, -1)$.

We then obtain the following plot:

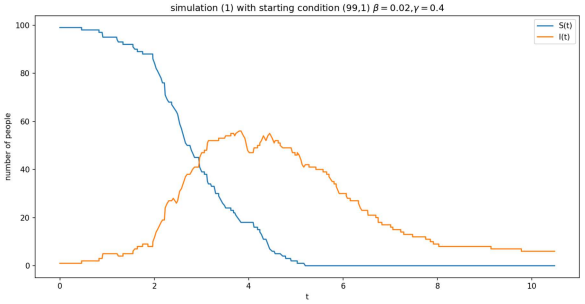


Figure 1: Simulation of the stochastic SIR model.

Mean field approximation of SIR:

When $N, S(t), I(t)$, are large, so that $(S(t), I(t)) + (-1, 1) \approx (S(t), I(t))$ and $(S(t), I(t)) + (0, -1) \approx (S(t), I(t))$, i.e., a single reaction changes very little the state, one can obtain the so-called mean field approximation of SIR which can be written in differential form as:

$$\frac{dS}{dt} = -\beta S(t)I(t),$$
$$\frac{dI}{dt} = \beta S(t)I(t) - \gamma I(t),$$

with $(S(0), I(0)) = (S_0, I_0)$, where we assume here $\forall t S(t), I(t) \in \mathbb{R}^+$.

This model is discretized using the forward Euler method with $N = 500$ time points and the time-step $h = \frac{T}{N}$ as:

$$S_j \approx S(j \cdot h)$$
$$I_j \approx I(j \cdot h)$$
$$S_{j+1} = S_j(1 - h\beta I_j)$$
$$I_{j+1} = I_j(1 + \beta h S_j - \gamma h),$$

for $j = 1, ..., N - 1$ and with $S_0 = 99$ and $I_0 = 1$.

We then obtain the following plot:

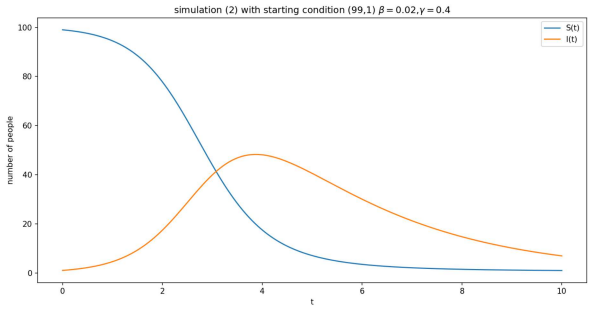


Figure 2: Simulation of the mean field approximation of SIR.

Diffusion approximation of SIR:

We now consider the following model:

$$dS = -\beta S(t)I(t)dt - \sqrt{\beta S(t)I(t)}dW_S$$

$$dI = (\beta S(t)I(t) - \gamma I(t))dt + \sqrt{\beta S(t)I(t)}dW_S - \sqrt{\gamma I(t)}dW_I,$$

with $(S(0), I(0)) = (S_0, I_0)$, and where W_S, W_I are two independent standard Wiener processes.

This model is discretized, using the Euler-Maruyama scheme introduced in the lecture notes, with $N = 500$ time points and the time-step $h = \frac{T}{N}$, by the following:

$$S_j \approx S(j \cdot h)$$

$$I_j \approx I(j \cdot h)$$
$$S_{j+1} = S_j(1 - h\beta)I_j - \sqrt{\beta S_j I_j}\xi_{1j}$$
$$I_{j+1} = I_j(1 + \beta h S_j - \gamma h) + \sqrt{\beta S_j I_j}\xi_{1j} - \sqrt{\gamma I_j}\xi_{2j},$$

for $j = 1, ..., N$, where $\xi_{1j}, \xi_{2j} \sim N(0, h)$ and with $S_0 = 99$ and $I_0 = 1$.

We then obtain the following plot:

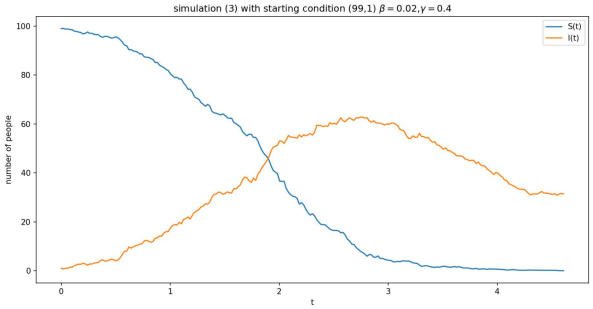


Figure 3: Simulation of the diffusion approximation of SIR

We observe on the plots that the dynamics generated by the three models are similar: the susceptible population starts at the value $S_0 = 99$ and decays to 0 rapidly, whereas the infected population starts at $I_0 = 1$, grows to a value in the range $[40, 60]$ and then decays to some value greater than zero. We can also note that the dynamics generated by the stochastic SIR and the diffusion approximation of SIR lead to curves with jumps, whereas the mean field approximation of SIR leads to smooth curves. This is given to the stochastic nature of the first two models, while the latter model is purely deterministic.

Exercise (b)

For $Z = \mathbb{1}_{\{I(t)=0 \text{ for some } t \leq T\}}$, we want to estimate $\mu = \mathbb{E}[Z] = \mathbb{P}(I(t) = 0 \text{ for some } t \leq T)$. We use a two step Monte Carlo method:

- Pilot run: run the stochastic SIR simulation with $\bar{N} = 20000$ replicas $Z^{(1)}, ..., Z^{(\bar{N})}$ and compute:

$$\hat{\mu}_{\bar{N}} = \frac{1}{\bar{N}} \sum_{i=1}^{\bar{N}} Z^{(i)}$$

$$\hat{\sigma}_{\bar{N}} = \frac{1}{\bar{N} - 1} \sum_{i=1}^{\bar{N}} (Z^{(i)} - \hat{\mu}_{\bar{N}})^2$$

- Set $N = \lceil \frac{c^2_{1-\frac{\alpha}{2}} \hat{\sigma}^2_{\bar{N}}}{\text{tol}^2} \rceil$
- Run the simulation with $(Z^{(1)}, ..., Z^{(N)})$.
- Compute

$$\hat{\mu}_N = \frac{1}{N} \sum_{i=1}^{N} Z^{(i)}$$

$$\hat{\sigma}_N = \frac{1}{N-1}\sum_{i=1}^{N}(Z^{(i)} - \hat{\mu}_N)^2$$

For $T = 1$, we obtain an estimate $\mu \approx 0.173$, a standard deviation $\sigma \approx 0.143$ and a confidence interval $[0.164, 0.182]$.
For $T = 2$, we obtain an estimate $\mu \approx 0.207$, a standard deviation $\sigma \approx 0.164$ and a confidence interval $[0.197, 0.217]$.
For $T = 10$, we obtain an estimate $\mu \approx 0.208$, a standard deviation $\sigma \approx 0.165$ and a confidence interval $[0.197, 0.218]$.

We can see that the probability of extinction of the disease approaches the value $\mathbb{P}(I(t) = 0) = 0.8$, for some $t \leq T$, when T grows.

Exercise (c)

We want to estimate the probability of extinction at $T = 2$ starting from the initial condition $(S_0, I_0) = (95, 5)$. In this setting, the case of extinction is a rare event. The Crude Monte Carlo estimator is very inefficient for rare events, as it needs a sufficiently large sample of occurrences of the event to be able to estimate its probability. In this case, Importance Sampling seems to be an adequate choice of a variance reduction technique to improve the convergence time of the estimator. Importance Sampling allows us to artificially increase the probability of the rare event, so that not as many evaluations are needed to see occurrences of the event.
In our specific case, we do this by increasing the probability for recovery, as it is the driving factor for an extinction of the disease in the SIR-model. We change the recovery rate $\gamma = 0.4$ to $\gamma_{proposal} = \gamma \cdot 5 = 2$ while keeping the infection rate $\beta = 0.02$ fixed. The multiplicative term 5 achieved good results for us. It is possible to take other multiplicative terms, but one should consider that multiplicative terms that are too low will not provide enough improvement; and multiplicative terms that are too high might not catch the probabilistic structure of the problem and might cause numerical instability in the calculation of the likelihood ratio w.

Results

The results can be seen in Table 1. The Crude Monte Carlo estimator (CMC) estimates an extinction probability of 0.0338%, while the Monte Carlo Importance Sampling estimator (MCIS) estimates an extinction probability of 0.0356%. Both values are included in each other's confidence intervals, so the difference in value is plausible. In the simulation, MCIS needs 46 times less evaluations than CMC and needs about 69 times less time to converge to a 5%-interval of the estimated probability. The stronger improvement in time compared to the improvement in evaluations can be explained by the following. In the MCIS a simulation with an early extinction of the pandemic happens more often. As soon as the pandemic is extinct, it will stay extinct until the end, so the simulation doesn't have to be carried out until time T. Hence, on average, a single evaluation of the process might take less time in MCIS than in CMC. In total, MCIS shows a significant improvement in convergence time compared to CMC.

	Extinction Probability	Variance	Confidence Interval	Evaluations	Time
MCIS	0.0356%	5.97×10^{-6}	$[0.0338\%, 0.0374\%]$	$66,694$	10.49 s
CMC	0.0338%	337.46×10^{-6}	$[0.0317\%, 0.0358\%]$	$3,071,897$	727.68 s

Table 1: Results of the Variance Reduction Technique.

Exercise (d)

Let $X(t) = (S(t), I(t), R(t))$. The stochastic SIR-d process associated to (4) is given by:

$$
\begin{aligned}
(a_1) &: \mathbb{P}(X(t+dt) = (s,i,r) + (1,0,0) | X(t) = (s,i,r)) = m(s+i+r)dt + o(dt) \\
(a_2) &: \mathbb{P}(X(t+dt) = (s,i,r) + (-1,0,0) | X(t) = (s,i,r)) = msdt + o(dt) \\
(a_3) &: \mathbb{P}(X(t+dt) = (s,i,r) + (0,-1,0) | X(t) = (s,i,r)) = (m+\upsilon)idt + o(dt) \\
(a_4) &: \mathbb{P}(X(t+dt) = (s,i,r) + (0,0,-1) | X(t) = (s,i,r)) = mrdt + o(dt) \\
(a_5) &: \mathbb{P}(X(t+dt) = (s,i,r) + (-1,1,0) | X(t) = (s,i,r)) = \beta sidt + o(dt) \\
(a_6) &: \mathbb{P}(X(t+dt) = (s,i,r) + (0,-1,1) | X(t) = (s,i,r)) = \gamma idt + o(dt),
\end{aligned}
$$

where (a_1) is the birth of a host, (a_2) the death of a susceptible host, (a_3) the death of a infected host, (a_4) the death of a recovered host, (a_5) an infection and (a_6) a recovery.

Exercise (e)

For the stochastic SIR d, we apply the following algorithm. Start with $X(0) = (S(0), I(0), R(0))$, and while $t < T$:

- Stop the simulation if $X(t) = (0,0,0)$.
- Generate an exponentially distributed holding time S, with parameter jump, with the jumping rate $\text{jump} = 2m(S(t) + I(t) + R(t)) + I(t)(\upsilon + \gamma + \beta S(t))$.
- Generate $U \sim U(0,1)$.
- Calculate the probabilities:

$$
\begin{aligned}
p_1 &= \frac{m(S(t) + I(t) + R(t))}{\text{jump}} \\
p_2 &= \frac{mS(t)}{\text{jump}} \\
p_3 &= \frac{(m+\upsilon)I(t)}{\text{jump}} \\
p_4 &= \frac{mR(t)}{\text{jump}} \\
p_5 &= \frac{\beta S(t) I(t)}{\text{jump}}
\end{aligned}
$$

- Let $k_i = \sum_{j=0}^{i} p_j$.

- If $U \leq k_0$, set $X(t) = X(t) + (1, 0, 0)$.
- If $k_0 \leq U \leq k_1$, set $X(t) = X(t) + (-1, 0, 0)$.
- If $k_1 \leq U \leq k_2$, set $X(t) = X(t) + (0, -1, 0)$.
- If $k_2 \leq U \leq k_3$, set $X(t) = X(t) + (0, 0, -1)$.
- If $k_3 \leq U \leq k_4$, set $X(t) = X(t) + (-1, 1, 0)$.
- Else set $X(t) = X(t) + (0, -1, 1)$.
- Set $t = t + S$.

The mean field approximation (4) is discretized using the forward Euler method with $N = 500$ time points and the time-step $h = \frac{T}{N}$ as:

$$S_j \approx S(j \cdot h)$$

$$I_j \approx I(j \cdot h)$$

$$R_j \approx R(j \cdot h)$$

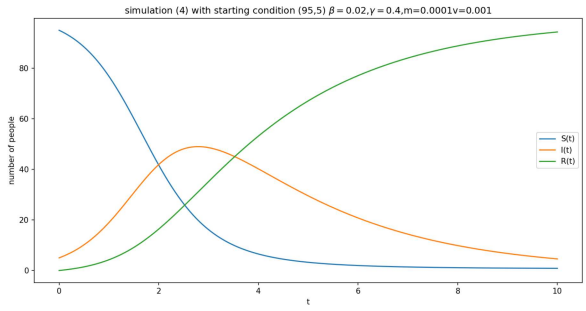
$$S_{j+1} = S_j(1 - h\beta I_j) + hm(S_j + I_j + R_j) - hmS_j$$

$$I_{j+1} = I_j(1 + \beta h S_j - \gamma h) - h(m + \upsilon)I_j$$

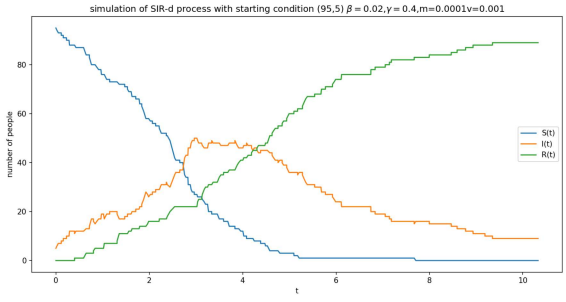
$$R_{j+1} = R_j + \gamma h I_j - hmR_j,$$

for $j = 1, ..., N - 1$ and with $S_0 = 95$, $I_0 = 5$ and $R_0 = 0$.

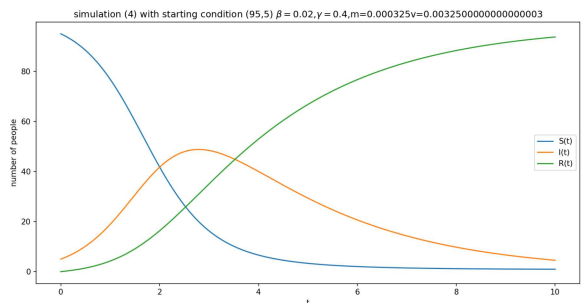
We then obtain the following plots:



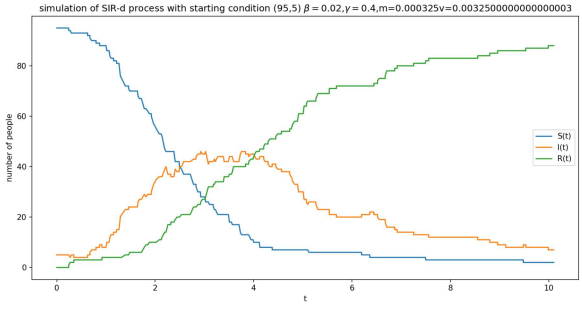
(a) $m = 0.0001, v = 0.001$.



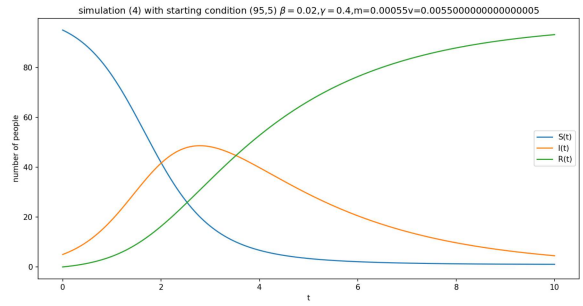
(b) $m = 0.0001, v = 0.001$.



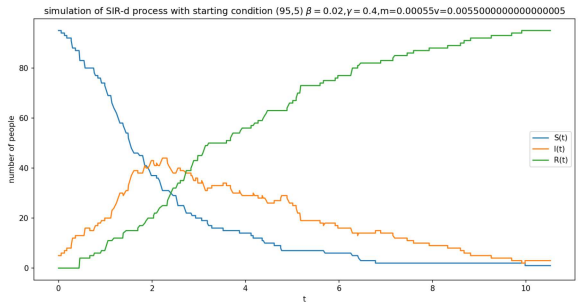
(a) $m = 0.000325, v = 0.00325$.



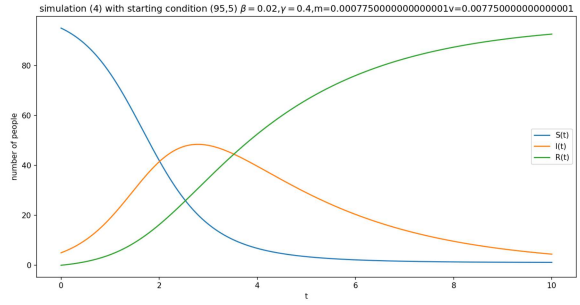
(b) $m = 0.000325, v = 0.00325$.



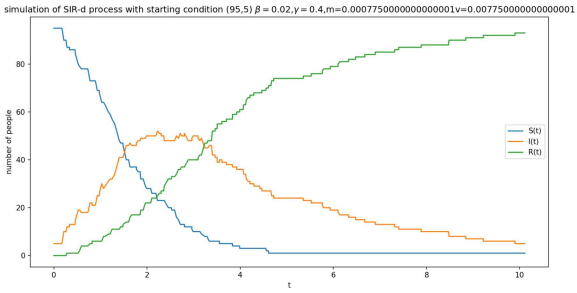
(a) $m = 0.0005, v = 0.005$.



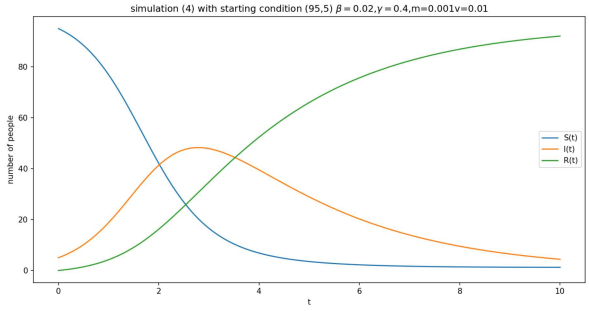
(b) $m = 0.0005, v = 0.005$.



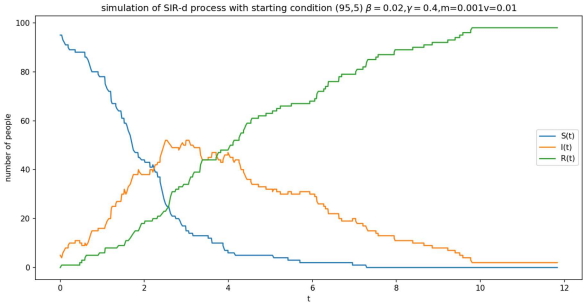
(a) $m = 0.0007, v = 0.007$.



(b) $m = 0.0007, v = 0.007$.



(a) $m = 0.001, v = 0.01$.



(b) $m = 0.001, v = 0.01$.

As we can see in the previous plots, the mean field approximation is not really affected by the fluctuation of the values of m and v. This could be explained by the fact that the mean field approximation is a deterministic model that describes the average behavior of the system using differential equations. In this case, it likely doesn't fully account for the stochastic effects and the influence of demographic fluctuations. On the other hand, the stochastic SIR-d process associated with (4) is more sensitive to the values of m and v, but it still behaves in the same global way for all the values of m and v.

Exercise (f)

For every β such that R_0 gets the values $\{1.01, 1.05, 1.1, 1.5\}$ namely, $\beta \in \{0.0041, 0.0043, 0.0045, 0.0061\}$ for the final times T $\in \{1, 2, 10\}$ we get the following estimator for the probability of extinction with the starting value X=(95,5,0):

Probability of extinction	T	R_0
0.01141198779021367	1	1.01
0.010789775093710954	1	1.05
0.010097764722081976	1	1.1
0.006432407267767804	1	1.5
0.052237300832685046	2	1.01
0.04987354577642893	2	1.05
0.04584089664909096	2	1.1
0.024370199922302244	2	1.5
0.45850914205344584	10	1.01
0.433728981206726	10	1.05
0.4021413276231263	10	1.1
0.1825780331722607	10	1.5

Table 2: Probability of extinction, T, and R_0.

As we can see in Table 2, as R_0 increases, the probability of extinction decreases for a given time T. This is consistent with the biological interpretation of R_0: higher R_0 values correspond to more transmissible infections, making extinction less likely.
However, the probability of extinction increases over time, even for higher R_0 values. This reflects the nature of stochastic processes, where random chance can eventually lead to the disease dying out, even if $R_0 > 1$, particularly in smaller populations.
To answer the question of whether the epidemic can go extinct at a time T for $R_0 = 1.01, 1.05, 1.1, 1.5$:

- For $T = 1$ and $T = 2$, the extinction probabilities are relatively low. Specifically, for $T = 1$, the probabilities lie between $[0.6\%, 1.1\%]$, while for $T = 2$, they range from $[2.4\%, 5.2\%]$. This suggests that extinction is unlikely for all values of R_0 within these timeframes, although it remains possible, with a very small probability.
- For $T = 10$, the probabilities of extinction increase significantly, ranging from $[18\%, 45\%]$. In this case, extinction has a non-negligible probability of occurring, especially for lower R_0 values.

In summary, while extinction is improbable for shorter timeframes (e.g., $T = 1, 2$) and higher R_0 values, it becomes much more likely over longer time horizons (e.g., $T = 10$), particularly for smaller R_0. These results highlight the role of stochastic effects in driving epidemic extinction, even when $R_0 > 1$.

Appendix

In the following we give the code for our models. Both the models file and evaluation file are in the submitted zip file.

Imports

```
#Stochastic Simulation of Epidemic Models
#Alexandre Francesco Corbelli, Luis Emilio Zaldivar Hanke, Victor Johan Svensson
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